

Association between a marine halacarid mite (Halacaroida: Arthropoda) and early juveniles of the northern sea cucumber *Cucumaria frondosa* in Nunavut

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Abstract

Juvenile and adult sea cucumbers are integral to marine food webs across the globe. The northern species *Cucumaria frondosa* is part of the diet of sea stars, crabs, fishes, birds and mammals, including seals and walruses, across its wide geographic distribution in the North Atlantic and Arctic Oceans. However, the trophic ecology of early juveniles is far less understood. In Qikiqtait (southern Nunavut, northern Canada), a nursery comprising newly settled and early juveniles <1 year of age (<1 mm in length) was recently identified. One juvenile sampled from this nursery was fortuitously discovered to host a transparent halacarid mite (~150 µm in length). The colour of the mite's digestive tract mirrored the colour of the sea cucumber's tissues, which are rich in red pigmented cells. Accordingly, the mite was cryptic, almost invisible when seen from above, and was presumed to feed on the tissues of juveniles. This represents the first observation of a direct interaction between juvenile sea cucumbers and a mite, providing novel insight on halacarid–echinoderm associations and adding an element to our understanding of northern sea cucumber populations.

Introduction

Juveniles and adults of the sea cucumber *Cucumaria frondosa* are known to be part of the diet of numerous cold-water and Arctic species. Among its major predators are sea stars, crabs, fishes, eider ducks, walruses, bearded seals and humans (Gianasi et al. 2021; Mercier et al. 2023). However, similar to all sea cucumbers, the ecology of early juveniles (<1 year) of *C. frondosa* in their natural environment is poorly studied, including their potential competitors and predators, as they are difficult to find. The recent discovery of a nursery of *C. frondosa* in Qikiqtait (Nunavut, Canada; Hamel et al. 2023 Nunavut, Canada) has unlocked exciting research perspectives on the ecology of early juvenile sea cucumbers.

Marine halacarid mites

The Halacaridae family (Arthropoda: Arachnida) consists of more than 1100 non-swimming species of mites that are distributed worldwide, from the intertidal zone down to abyssal depths (Bartsch 2009). Halacarids are mostly benthic marine species, but some have adapted to freshwater. They are found on a variety of substrata, including macroalgae, muddy and sandy bottoms, sponges, hydrozoans, bryozoans, barnacles, mussels and polychaetes (Bartsch 2006). Twenty halacarid species, belonging to 10 genera, have been described living in the same environments or in association with all five extant classes of echinoderms (Chatterjee 2021). However, the exact nature of the relationship between halacarid mites and echinoderms is poorly understood because live observations are rare (Chatterjee 2020). For instance, one species of halacarid was described

as a parasite in the gut of a sea urchin in Indonesia (Viets 1938) and another was observed crawling on a brittle star in Australia (Bartsch 1993). Four more halacarid species were reported to be associated with sea cucumbers in the Adriatic Sea (Viets 1940), without providing any details, and another six species were found in sediments where *Cucumaria* sp. co-occurred in Norway (Viets 1928). Because they have never been directly seen together *in situ*, the relationship between halacarid mites and sea cucumbers remains an enigma.

In this study, recently settled juveniles of *Cucumaria frondosa* (~3 months old) were collected in the subtidal zone (~1–4 m depth) of the Kataaluk channel north of Renouf Island (Qikiqtait, Nunavut, Canada; 56.603611, -79.142500). Stones of different sizes from 1 cm to 36 cm in diameter were collected with a long pole net, brought to the boat and examined for juvenile sea cucumbers. Several hundred juvenile sea cucumbers (≤10 mm long) were collected from the surfaces or crevices of stones. They were stored in Eppendorf tubes with seawater and transported to the laboratory for further study and processing.

While sorting live juveniles (which are orange-reddish) under the microscope, one was found to harbour a transparent halacarid mite on the surface of its body. The mite (~150 µm in length) was seen walking on the juvenile (~900 µm) for five minutes, exploring the external body wall with its palps and inserting chelicerae into it (Fig. 1A, B). The body and digestive tract of the mite clearly showed the pale reddish colour of the tissues of the juvenile sea cucumber, suggesting that it ingested them. The mite most probably consumes the fluids containing red pigmented cells, chiefly coelomocytes or their aggregates, as described in Caulier et

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al. (2020). This would be achieved by piercing the body-wall epithelium to reach the hydrovascular system of the juvenile sea cucumber. In support of this assumption, some mites have been described sucking the fluid from their prey using specialised piercing mouthparts (Dabert 2005). Considering the size of the mites, they may solely target early juveniles of *C. frondosa* that possess a thin body wall. The possibility that they can also pierce the thicker epithelia of adults, including that of the podia, would need to be investigated as mites were found in the same habitat as adults of the species in Norway (Viets 1928).

Given that mites are small relative to juvenile sea cucumbers (ratio of 1:6 in the current study; down to 1:66 for juveniles up to 10 mm), it is unlikely that they represent a major burden, although intense or repetitive predation, especially on smaller individuals (≤ 2 mm), may impede their growth rate. While the collection method was not well adapted to the preservation of free-living epibionts (see discussion below), the density of mites seems to be low in comparison to the density of small juvenile sea cucumbers (up to 104 ind/m², Hamel et al. 2023), further spreading the potential burden they represent. Video observations clearly showed the rapid displacement of the mite on the surface of juvenile sea cucumbers, supporting its capacity to quickly move from one host or prey to another. However, further investigations will be required to determine the abundance of halacarid mites and their potential impacts on early juvenile sea cucumbers inside the nursery, as most of them may have been lost during the sampling. The long pole net scraped the bottom and brought the rocks to the surface, during which time the mites could have detached and fallen off. Moreover, the early juvenile sea cucumbers were collected from the rocks using tweezers and the mites may have been accidentally crushed. Finally, the mites are small and well camouflaged, and nearly invisible to the naked eye. During this field expedition in Qikiqtait, a second mite was fortuitously observed crawling on another juvenile sea cucumber sampled in the intertidal zone near the hamlet of Sanikiluaq, while a third one was found under a rock (in a sea cucumber habitat), suggesting that more mites might be discovered with further and more careful sampling.

Interestingly, the color of the mite matched almost perfectly the color of the sea cucumber host, making it almost invisible when viewed from above (Fig. 1C). This suggests a mimetic adaptation, likely to prevent detection by predators, and supports the possibility that the mite concentrates on juvenile sea cucumbers as a food source. The observation of mimesis and evidence of feeding on juvenile sea cucumbers suggest a parasitic relationship between the two species, to the advantage of the mite.

This contribution represents a first direct observation of a mite crawling and feeding on a juvenile sea cucumber, possibly the first association between living Halacaridae and Holothuroidea as a whole. The observation also, quite possibly, represents a newly described, non-lethal predator of *C. frondosa*. As fishing pressure on *C. frondosa* continues to increase across its distribution range, and with potential

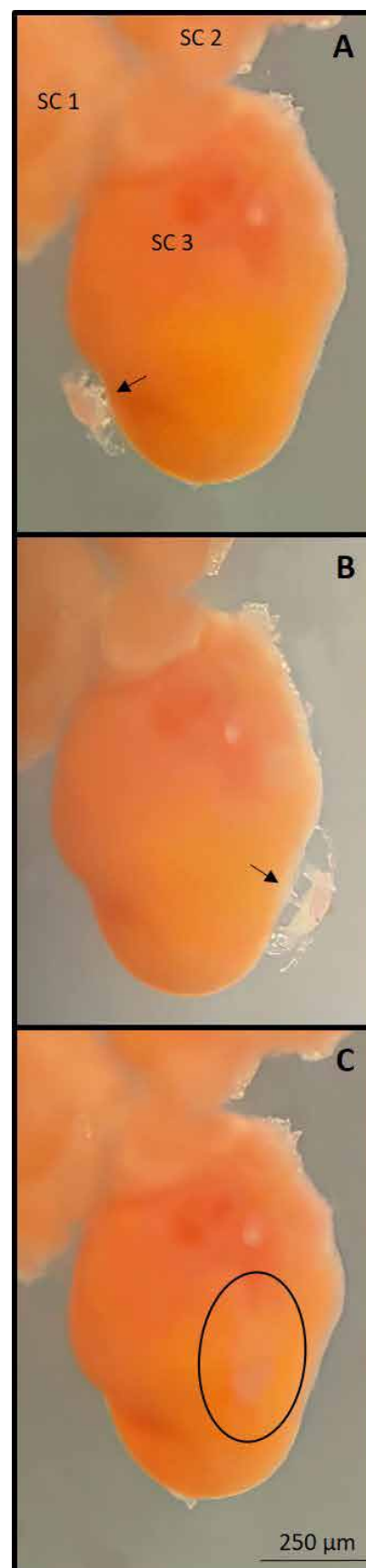


Figure 1. Still frames extracted from video footage showing the mite (arrow pointing at it) actively moving over the juvenile *Cucumaria frondosa*. The reddish colour of the digestive tract of the mite rendered it nearly invisible from certain angles (circle), which is in line with the ingestion of sea cucumber tissues.

SC: sea cucumber.

expansion of small-scale fisheries into the Canadian Arctic in the near future, close monitoring of nursery habitats and of the trophic position of early life stages in food webs should be undertaken to ensure sustainable activities.

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